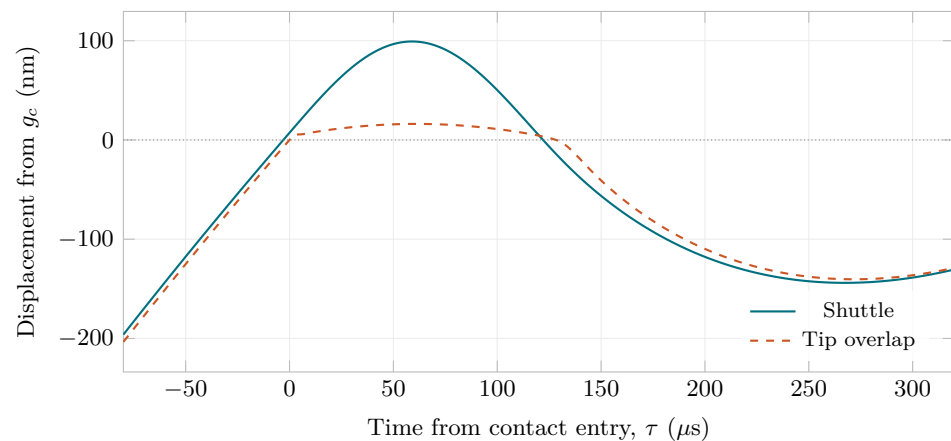


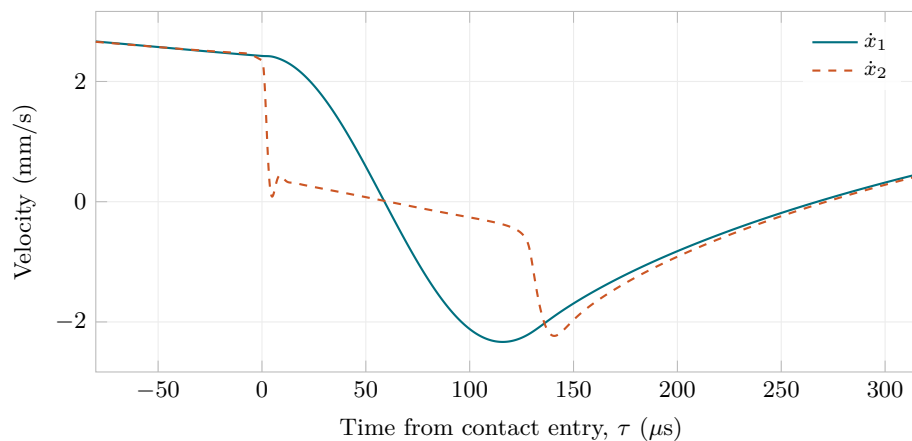
What happens when an electrode tip touches?

First positive-wall entry in the final drive cycle $\tau = 0$: $x_2 = g_c$ $0.1 \mu\text{s}$ sampling

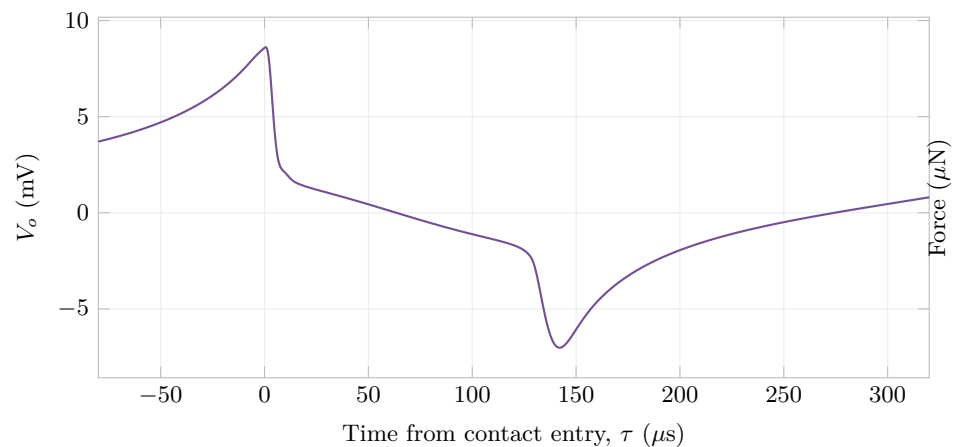
(a) Motion relative to the contact position



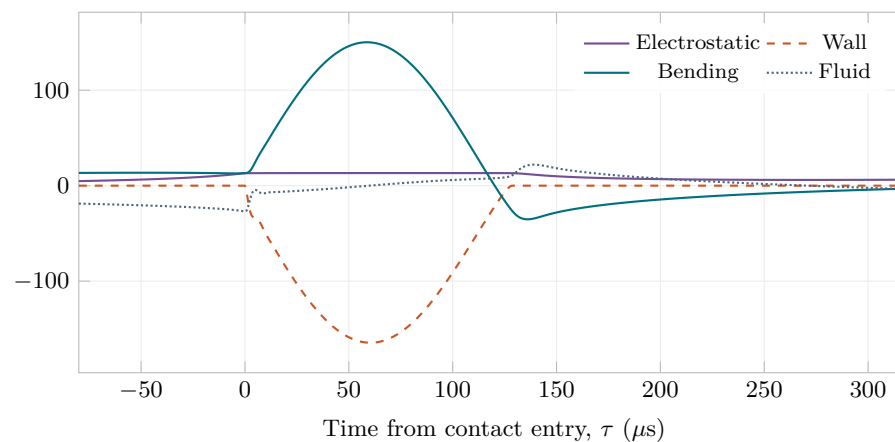
(b) Mechanical velocities



(c) Contact-scale electrical signal



(d) Generalized forces on the tip coordinate



Positive overlap is permitted by compliant contact. The tip can slow while the shuttle continues moving, bending the electrode.

Forces in (d) are signed, summed over all mobile beams, and conjugate to x_2 . Base inertia and cross-coordinate inertia are not plotted in that panel.